

SCHOOL OF COMPUTER SCIENCE AND IT YBN UNIVERSITY RANCHI

JAVA PROGRAMMING ASSIGNMENT 06

Java Switch Statements

1) Simple Calculator Using Switch

Description: Write a Java program that functions as a simple calculator. The program should prompt the user to input two numbers and an operator (+, -, *, /) and use a switch statement to perform the appropriate operation.

Hint:

- Use the switch statement to evaluate the operator and perform the corresponding arithmetic operation.

2) Day of the Week

Description: Write a Java program that takes an integer input from the user (1-7) and uses a switch statement to print the corresponding day of the week (e.g., 1 for Monday, 2 for Tuesday, etc.).

Hint:

- Use the switch statement to map each number to its corresponding day.

Java Break

3) Find First Divisible Number

Description: Write a Java program that iterates over numbers from 1 to 100 and finds the first number divisible by both 3 and 7. Use a break statement to exit the loop once the number is found.

Hint:

- Use a for loop to iterate and if conditions to check divisibility. Use break to stop when the condition is met.

4) Search for a Specific Character in a String

Description: Write a Java program that prompts the user to input a string and a character to search for. The program should use a loop to search for the first occurrence of the character in the string and use break to exit the loop once found.

Hint:

- Use a for loop to iterate over the string and if to check for the character. Use break to exit once the character is found.

Java Continue**5) Skip Multiples of 5**

Description: Write a Java program that prints numbers from 1 to 50 but skips multiples of 5 using the continue statement.

Hint:

- Use a for loop to iterate through the numbers and if to check for multiples of 5. Use continue to skip printing those numbers.

6) Print All Characters Except Vowels

Description: Write a Java program that prompts the user to input a string and prints all characters except vowels. Use the continue statement to skip vowels.

Hint:

- Use a for loop to iterate over the string and if to check for vowels. Use continue to skip printing vowels.

Break and Continue in While Loop**7) Find the Sum of Numbers Until 100 is Reached**

Description: Write a Java program that prompts the user to input numbers continuously and adds them to a sum. Use a while loop, and if the sum reaches or exceeds 100, use break to stop the loop.

Hint:

- Use a while loop to repeatedly take input and add to the sum. Use break when the sum reaches 100.

8) Skip Negative Numbers in Input

Description: Write a Java program that prompts the user to input numbers continuously. The program should sum only positive numbers and use continue to skip any negative numbers. Exit the loop if the user enters 0.

Hint:

- Use a while loop for continuous input. Use continue to skip negatives and break to exit on 0.

Break and Continue in for Loop

9) Find the First Prime Number in a Range

Description: Write a Java program that finds the first prime number in a range (1-100) using a for loop. Use break to exit the loop once the first prime number is found.

Hint:

- Use a for loop to iterate through the range and nested loops to check if a number is prime. Use break when the first prime is found.

10) Skip Printing Odd Numbers

Description: Write a Java program that prints all even numbers between 1 and 50 using a for loop and the continue statement to skip odd numbers.

Hint:

- Use a for loop to iterate from 1 to 50 and continue to skip printing odd numbers.

These problems cover various aspects of control flow in Java, including switch statements, break, continue, and their application in different types of loops.